

Optimization of charging strategy for lithium-ion battery packs based on complete battery pack model

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ABSTRACT

This study focuses on a charging strategy for battery packs, as battery pack charge control is crucial for battery management system. First, a single-battery model based on electrothermal aging coupling is proposed; subsequently, a battery pack cooling model and battery pack equilibrium management model are combined to form a complete battery pack model that describes the state parameters of the battery pack and the interaction between the individual batteries in the battery pack. Then, a multi-objective optimal charging strategy considering charging time, aging, and energy loss is proposed, and the equilibrium management, temperature, and battery parameters are restricted. A genetic algorithm method was used to optimize the adaptive multi-phase constant-current constant-voltage charging strategy. A fast charging strategy based on the shortest charging time is proposed. The results show that the fast charging strategy can significantly reduce charging time but leads to aging and increases energy loss. The contradictory relation between charging time and aging and the interaction between charging time and energy loss are revealed. The balance optimization charging strategy appears on the left side of the Pareto curve. Finally, a balanced charging strategy considering charging time, aging, and energy loss is obtained. In comparison with single batteries with the same average initial current charging, the constant current phase order decreases because of battery pack state parameters and equilibrium management, resulting in different charging strategies.

1. Introduction

In recent years, with increasing attention to environmental issues in various countries, electric vehicles have rapidly developed on a global scale. Lithium-ion batteries have attracted considerable attention because they are used as the power core of electric vehicles. However, as the voltage and capacity of a single lithium battery are limited by its characteristics, satisfying the power and endurance requirements of electric vehicles is difficult. Generally, hundreds or thousands of batteries can be connected in series and parallel [1] to build battery packs to provide the voltage needed by electric vehicles and ensure cruising ability. However, owing to the inconsistency of the battery when it leaves the factory, it will lead to over discharge or overcharge of the battery pack. State of charge (SOC) estimation, dynamic monitoring, and equilibrium management of battery packs are usually performed by the battery management system. A considerably longer charging time

will reduce the utility of electric vehicles and seriously affect user experience [2]. A high charging current leads to battery capacity attenuation and even safety problems [3].

The literature [4] summarizes the charging strategies of commercial lithium-ion batteries and indicates that the passive charging strategy (CCCV [5]) is simple to implement but lacks the ability to maintain good robustness. An active charging strategy can effectively improve the performance and efficiency of the battery. In the literature, various active charging strategies with different optimization objectives have been reported. In [6], the dynamic horizon estimation algorithm is adopted to monitor the internal state information of the battery, which can greatly shorten the charging time while satisfying health-related constraints. In [7], a charging optimization strategy based on a dynamic programming algorithm is proposed to target charging time and battery aging. In [8], a charging algorithm based on LTV-MPC was proposed, and the optimal charging curve was obtained in the form of CC-CV-CT. In [9], a charging strategy is proposed to reduce the

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List of symbols

SOC	State of charge
CC	Constant current
CV	Constant voltage
CT	Constant temperature
LTV-MPC	Linear time varying-Model Predictive Control
NSGA-II	Non-Dominated Sorted Genetic Algorithm-II
VRP	voltage relaxation curve
SEI	solid electrolyte interphase
BTMS	Battery thermal management system
EOL	End of life
SOH	State of health
PWM	Pulse Width Modulation

charging loss of lithium-ion batteries. The proposed charging strategy utilizes adaptive current distribution based on the internal resistance of the battery changing with the charging state and rate. In [10], a constant temperature and constant-voltage charging technology was proposed. A simple and easily usable proportional integral derivative controller is adopted in the charging scheme, and the charging current is dynamically adjusted according to the battery temperature. In [11], economic indices of total charging cost caused by battery aging and power loss are considered, and NSGA-II is used to optimize multistage constant-current charging strategy. In [12], a sinusoidal current is introduced in the charging process to reduce aging. In [13], the lithium plating process by online analysis of VRP was detected to adjust the charging current and mitigate battery aging. In [14], two optimal charging curves are used alternately to extend the cycle life. In one charging mode, the circulating capacity loss is small, but the charging capacity is reduced; in the other charging mode, the circulating capacity loss is large, but the charging capacity is increased. In [15], an optimal charging strategy to reduce the temperature increment and energy loss while charging the battery was proposed. In [16], based on the physical battery model, the authors study the charging strategy of optimal tradeoff between temperature rise, charging time, and loss. In [17], a real-time charging control framework proposed based on explicit MPC, exploiting its advantage in characterizing an explicit solution to an MPC problem, to enable real-time charging control. In [18], a first-order SEI layer growth rate model was introduced into the charging model and the nonlinear MPC method was introduced to solve the challenge of minimizing the SEI layer growth in the charging process. In [19], reduced order electrochemical life models are developed, and an optimal charging current at different SOC has been found using nonlinear MPC, which reduces side reaction rate and lithium plating rate. In [20], A semi-optimal charging protocol which can automatically modify the charging current is developed to enable the battery to be charged at near the maximum current without lithium plating.

The inconsistencies of a single battery cannot be eliminated, and the battery pack cannot simply be considered as a high-voltage, large-capacity battery. Several single-battery charging strategies are difficult to apply to battery pack charging. In addition, the complex circuit model in the battery pack and the interaction between the various batteries (including the influence of electricity and heat) make it difficult to model the battery pack. The thermoelectric model of battery pack has been established in many literatures. In [21], establishing a battery pack model for electrothermal coupling to discuss the influence of the number of parallel batteries on load imbalance. In [22], the authors coupled the flow resistance network model with the transient heat transfer model to calculate the influence of air flow inside the battery pack on the temperature distribution of the battery pack. In [23], The thermal equivalent circuit model of battery pack describing the thermal behavior of battery modules is established, which has a good agreement with those

from experiment and the maximum deviation is below 5%. In [24], a thermal management system equipped with micro heat pipe array of battery pack is designed. An equivalent thermal resistance model is developed for micro heat pipe array based on thermal circuit method. Which can make the maximum temperature difference maintain below 2 °C. In [25], the authors capture battery pack dynamics and maintain a simple topology by adding moving average noise to the single-resistor-capacitor circuit model. In [26], Each cell in battery pack is explicitly modelled as the Single Particle Model with electrolyte and thermal dynamics. And a nonlinear MPC is applied to optimally charging the battery pack. In [27], The flow resistance network model and heat transfer model are introduced to calculate the velocities of the cooling channels and battery cell temperature. the configuration of the battery pack in parallel air-cooled BTMS is optimized through arranging the spacings among the battery cells for cooling performance improvement. In [28], The literature proposes the manipulation of balancing current, reserved for charge or voltage balancing in battery pack, to additionally reduce temperature heterogeneity. And a MPC scheme is used to determine balancing current. A splicing equivalent circuit model is therein proposed and established [29]. Through an equivalent simulation of different internal effects in the charging and discharging process of the lithium-ion battery, the proposed model realizes an accurate mathematical expression of the special working condition and working process of the lithium-ion battery. In [30], a control-oriented battery pack model was proposed that describes the propagation process of aging and its impact on battery life. In addition, to reduce the impact of inconsistencies between batteries, a previous study [31] proposed an online equalization algorithm for lithium-ion battery packs based on rechargeable battery voltage curves. In [32], a hierarchical balancing circuit topology based on the Buck-Boost module was proposed. In [33], an active battery self-balancing algorithm based on SOC estimation of 24 tandem li-ion batteries was proposed. In [34], a combined equalization method based on multi-winding transformer was proposed. A non-dissipative equalization scheme based on fuzzy logic control [35] has been proposed to address the inconsistencies of series-connected lithium-ion batteries.

Although the above strategies have made some progress, Many active charging strategies have complex charging curves and higher control requirements. And not suitable for existing commercial charging equipment. In addition, describing the complete model of battery pack is still a problem to be solved (including the thermoelectric aging model, cooling model, and equilibrium model of battery pack). In addition, several studies on the battery pack charging strategy have focused on equilibrium management, but the optimal charging mode for battery packs has not been discussed yet. Compared with existing reports, our study makes three major contributions. First, a complete battery pack model is established for series-connected battery packs; this model includes a battery cell model of electrothermal aging coupling, A battery pack cooling model used to describe the thermal inconsistency of cells in a battery pack, and a battery pack equilibrium model used to describe battery electrical inconsistencies in a battery pack. Moreover, an adaptive multi-objective optimization charging strategy for battery packs is proposed. This charging strategy can be realized without complex controller, and the existing charging pile is relatively reformed. strategy involves charging time, aging, and energy loss as optimization objectives. At the model level, the battery equilibrium, temperature, and battery parameters are restricted to establish a complete multi-objective optimization mathematical model. Finally, the charging current distribution under different conditions is solved by adjusting the weight coefficient. The shortest charging time and balanced charging strategy are obtained, and the relation between charging time and aging and the interaction between charging time and energy loss are analyzed.

The remainder of this paper is organized as follows. The optimized model of the coupled thermoelectric-aging battery is presented in Section 2. The complete battery pack model is described in Section 3. The control strategy of the battery pack charging system is presented in

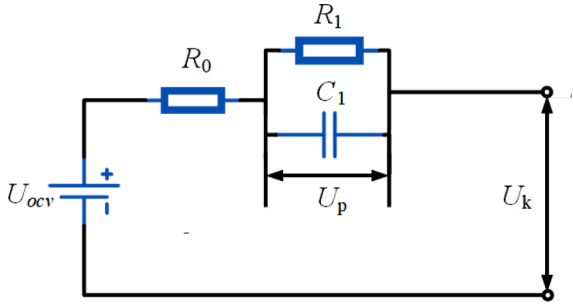


Fig. 1. First-order RC equivalent circuit model.

Section 4. The relevant simulation results are presented in Section 5, followed by concluding statements in Section 6.

2. Battery cell model

2.1. Electrical model

The electric model of the battery adopts the first-order resistor–capacitor (RC) equivalent circuit model as shown in Fig. 1. This includes open-circuit voltage (OCV), an RC pair (R_1 , C_1), and a resistor (R_0). In a cell, OCV is defined as the voltage measured using a voltmeter at both ends of the cell when no current passes through the cell. R_1 and C_1 denote the polarization resistance and polarization capacitance, respectively, which are used to describe the transient reaction caused by polarization and diffusion effect. The corresponding equation can be expressed as

$$\frac{dSOC}{dt}(t) = \frac{I(t)}{C_{bat}} \quad (1)$$

$$\frac{dU_p}{dt}(t) = -\frac{U_p(t)}{R_1 C_1} + \frac{I(t)}{C_1} \quad (2)$$

$$U_k(t) = U_{ocv} + U_p + I(t)R_0 \quad (3)$$

where C_{bat} denotes the rated capacity, $I(t)$ is the current, and $U_k(t)$ is the terminal voltage. The voltage of SOC and RC pairs (U_p) has been successfully identified in [36] through appropriate experimental design, and the results are shown in Fig. 2.

2.2. Thermal model

Fig. 3 is a schematic of the battery thermal model. In the battery thermal sub-model, heat generation and temperature in the battery are assumed to be uniformly distributed [37]. The relation between the internal heat generation and external heat dissipation of the battery is expressed as follows:

$$mc \frac{dT_c}{d\tau} = Q_0 - Q_c \quad (4)$$

where Q_0 denotes a single cell generates heat, and Q_c is the heat absorbed from the cell by the cooling channel. Heat generation [38] in the battery mainly includes the irreversible heat generated by the battery resistance (ohmic internal resistance and polarization internal resistance) and the reversible heat generated by the entropy change caused by the lithium-ion disintercalation and intercalation in the anode and cathode. The heat produced can be expressed as follows:

$$Q_0 = Q_{ohm} + Q_p + Q_r \quad (5)$$

$$Q_{ohm} = I^2 R_0 \quad (6)$$

$$Q_p = I^2 R_1 \quad (7)$$

$$Q_r = IT \left(\frac{dU_{ocv}}{dT_c} \right)_{soc} \quad (8)$$

where Q_{ohm} denotes ohmic resistance heat generation, Q_p is polarization heat generation, and Q_r is reversible heat. dU/dT_c is the temperature rise

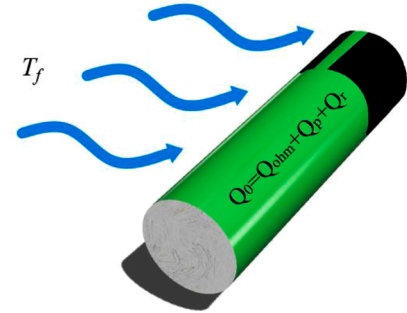


Fig. 3. Schematic of the thermal model.

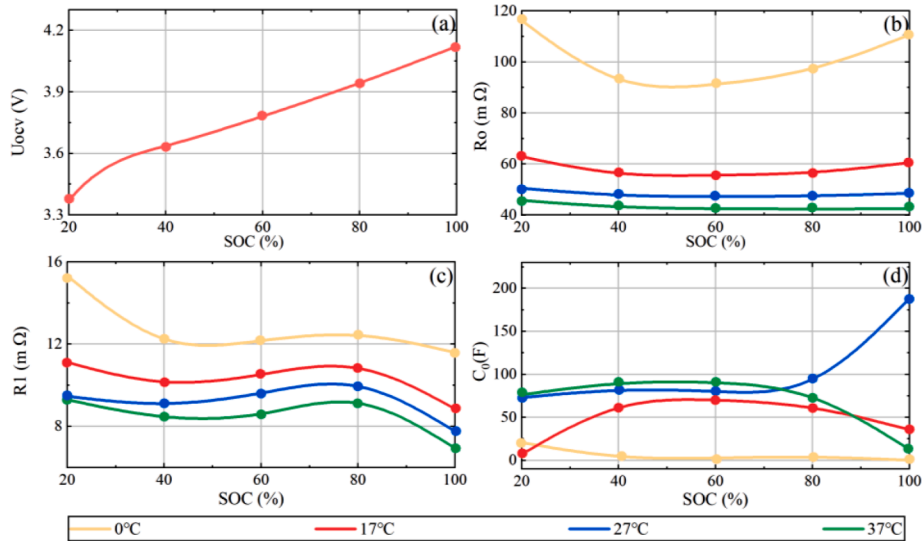
Fig. 2. Electrical parameters under different temperatures and SOC: (a) U_{ocv} , (b) R_0 , (c) R_1 , and (d) C_1 .

Table 1
Entropic coefficient of electromotive force.

SOC/%	0	0.2	0.4	0.6	0.8	1
dU/dT_c /(mV/K)	4.80	2.30	0.80	0.28	0.24	0.2

Table 2
B values under different discharging rates.

c-rate	0.5	2	6
B	31,630	21,681	12,934

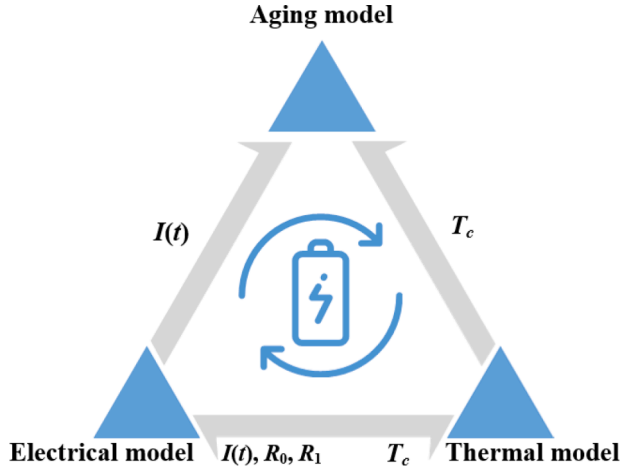


Fig. 4. Thermoelectric-aging coupling model.

coefficient of electromotive force. Table 1 shows the value under different SOCs; the convective heat transfer between the battery and the surrounding fluid can be calculated based on the following formula:

$$Q_c = hA(T_c - T_f) \quad (9)$$

where A is the heat transfer area on the battery surface, T_f is the external fluid temperature, and h is the convective heat transfer thermal resistance.

2.3. Aging model

Capacity attenuation is described as a function of discharge rate, temperature, and discharge depth [39]. The aging model was established through multiple orthogonal experiments. The temperature range

was -30 – 60 °C, the discharge depth was 10%–90%. The semi-empirical formula for battery aging is

$$Q_{loss} = Be^{\frac{-31700+370.3C}{8.3147}}(Ah)^{0.55} \quad (10)$$

where B is the constant coefficient under the influence of discharge multiplier. Table 2 provides the value of coefficient B under different discharge multipliers and performs polynomial fitting on it to obtain the expression formula of B with discharge multiplier C, Ah is the throughput of current in the corresponding time, and C is the discharge multiplier.

$$B(C) = -47.836C^3 + 1215C^2 - 9418.9C + 36042 \quad (11)$$

It is generally believed that the EOL of lithium-ion batteries is exhausted when the battery capacity loss reaches 20%, that is, when Q_{loss} is 20%, the SOH is 0. When the battery is new, Q_{loss} is equal to 0, and the corresponding SOH₀ is 1. Accordingly, SOH can be expressed as

$$SOH = 1 - Q_{loss} / 20\% \quad (12)$$

2.4. Electrothermal aging model

The high-fidelity single-cell model can be coupled with the electrothermal aging model. The electrical model represents the battery's electrical behavior, including current, voltage, SOC, among others, and influences the battery thermal model. The thermal model captures the thermogenic behavior of the battery and its relation with external heat dissipation and feeds the temperature back to the electrical model for parameter modification. The aging model can show the capacity loss of the battery during the cycle, which is affected by both the electrical model and the thermal model. These sub-models are interrelated and affect each other. The relation diagram is shown in Fig. 4.

3. Battery pack model

3.1. Electrothermal aging model for series-connected battery pack

The battery pack model is composed of 12 single cells in series. The monomer battery electric heating aging model has been built in the second. As the individual characteristics of the battery parameters can cause distortion in the battery pack model if a simple monomer battery model is extended to the battery pack, we need to modify the model of the battery pack definition. According to Kirchhoff's law, in series with the battery voltage and current, we have

$$U_{k,pack}(t) = \sum_{1 \leq i \leq 12} U_{k(i)}(t) \quad (13)$$

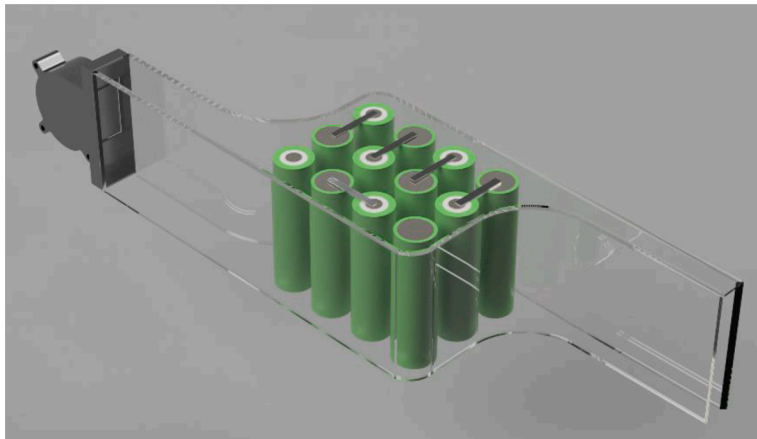


Fig. 5. Series air-cooled battery pack.

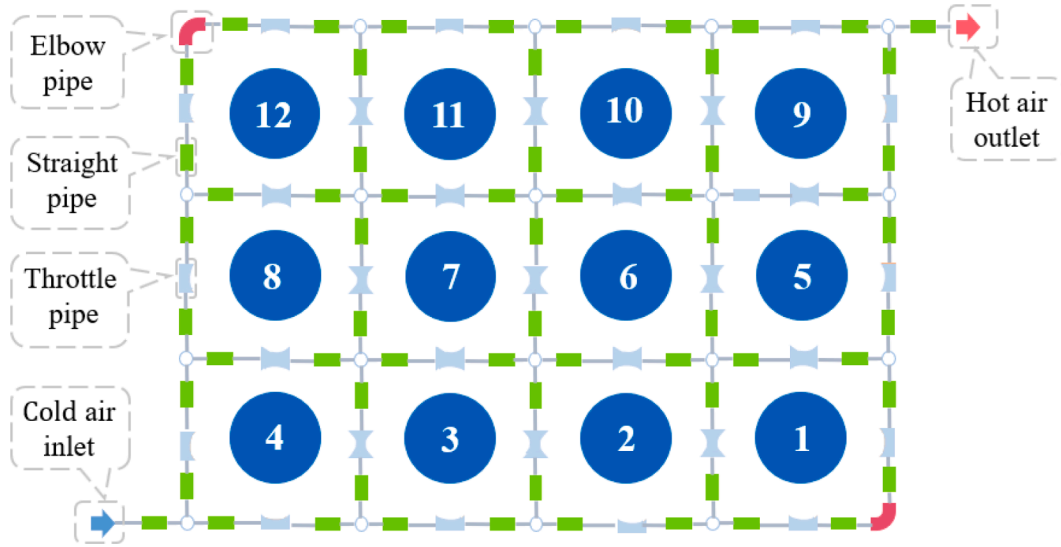


Fig. 6. Schematic view of battery pack's cooling model.

$$I_{pack}(t) = I_i(t) \quad (14)$$

where $U_{k,pack}(t)$ is the battery pack terminal voltage, $U_{k(i)}$ is the cell terminal voltage, $I_{pack}(t)$ is the battery pack current, and $I_i(t)$ is the cell current. For dissipation unit equalization, units with the minimum battery capacity are fully charged and discharged. Therefore, the packaging capacity is the minimum battery capacity [40]:

$$C_{pack}(t) = \min_{1 \leq i \leq 12} C_i(t) \quad (15)$$

where $C_{pack}(t)$ is the capacity of the battery pack, and $\min_{1 \leq i \leq 12} C_i(t)$ is the capacity of the single battery with the minimum capacity. The SOC of a battery pack is defined as the ratio of the minimum available capacity of a single battery to the rated capacity of the battery pack:

$$SOC_{pack}(t) = \frac{\min_{1 \leq i \leq 12} SOC_i(t) C_i(t)}{C_{pack}(t)} \quad (16)$$

where $SOC_{pack}(t)$ is the SOC of the battery pack and $\min_{1 \leq i \leq 12} SOC_i(t) C_i(t)$ is the minimum available capacity of the current single battery. The SOH concept of a single battery is extended to the battery-pack level. The capacity of the package is defined as the ratio between the minimum capacity of a single battery after cyclic charging and discharging and the rated capacity of the initial battery pack.

$$SOH_{pack}(t) = \frac{\min_{1 \leq i \leq 12} SOH_i(t) C_i(t)}{C_{pack}(t)} \quad (17)$$

where $SOH_{pack}(t)$ is the SOH of the battery pack, and $\min_{1 \leq i \leq 12} SOH_i(t) C_i(t)$ is the minimum capacity of a single battery after cyclic charging and discharging.

3.2. Battery pack thermal model

Due to the uneven cooling caused by the different environment of the battery in the battery pack, a large temperature gradient appears among the batteries, further aggravating the inconsistency of single batteries in the battery pack. Therefore, establishing the thermal model of the battery pack is crucial. The battery pack air cooling model established herein is shown in Fig. 5, and its flow field power is provided by an external fan. The battery pack heat model includes the flow resistance network model and the battery heat generation model. The flow resistance network model is used to simulate the flow field of the air-cooling system and provides the convective heat transfer coefficient of the air interface of the battery unit.

The inlet and outlet pipe sections are equivalent to the circular pipe with the corresponding hydraulic diameter, and the air duct around the battery is equivalent to the straight pipe, elbow pipe, and orifice plate with the corresponding size so as to realize the calculation of pressure loss and heat exchange of different air ducts. The equivalent figure is shown in Fig. 6. A previous study [41] formulates the flow resistance network model. The energy conservation equation of a single battery in the battery pack can be obtained by combining the heat generation model of the battery:

$$mc \frac{dT_{c,i}}{d\tau} = Q_{0,i} - \sum_{n=1}^4 Q_{c,i,n} \quad (18)$$

$$Q_{c,i,n} = h_{i,n} A_{i,n} (T_{c,i} - T_{f,i}) \quad (19)$$

$$h_{i,n} = \frac{Nu_{i,n} \lambda}{l_{i,n}} \quad (20)$$

$$Nu_{i,n}^{[38]} = \begin{cases} \max \left[\left(1.86 \frac{Re_{i,n} \cdot Pr \cdot d_{i,n}}{l_{i,n}} \right)^{\frac{1}{3}} \cdot \left(\frac{\mu}{\mu_s} \right)^{0.14}, \left(1.86 \frac{Re_{i,n} \cdot Pr \cdot d_{i,n}}{l_{i,n}} \right)^{\frac{1}{3}} \cdot \left(\frac{\mu}{\mu_s} \right)^{0.366} \right] & Re_{i,n} < 2300 \\ 0.027 Re_{i,n}^{0.8} \cdot Pr^{\frac{1}{3}} \cdot \left(\frac{\mu}{\mu_s} \right)^{0.14} & Re_{i,n} > 2300 \end{cases} \quad (21)$$

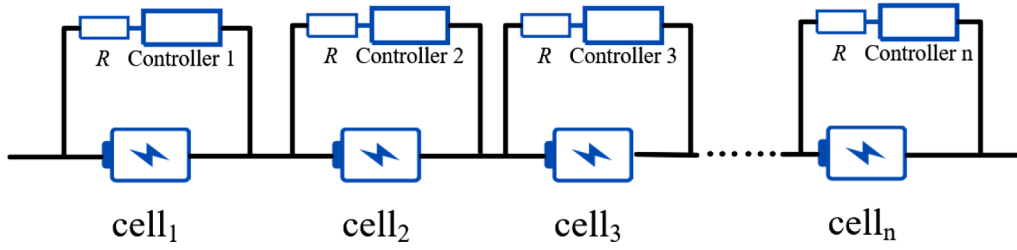


Fig. 7. Block diagram of resistor equalizing circuit.

Table 3

Cell's capacity and Initial SOC in the battery pack.

	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6	Units
Capacity	3.28	3.31	3.29	3.33	3.25	3.35	Ah
Initial SOC	9.15	10.25	11.00	10.54	10.23	9.97	%
	Cell 7	Cell 8	Cell 9	Cell 10	Cell 11	Cell 12	Units
Capacity	3.31	3.30	3.31	3.32	3.27	3.35	Ah
Initial SOC	9.23	9.42	9.29	10.60	10.04	9.79	%

where d is hydraulic diameter, l is characteristic length, subscript i is the i th battery cell and n is the heat transfer surface around the cell.

The logarithmic mean temperature difference model is adopted for the temperature difference between the battery and the air:

$$\Delta T_{i,n} = (T_{f,n} - T_{f,in}) \left/ \ln \frac{T_{c,i} - T_{f,in}}{T_{c,i} - T_{f,out}} \right. \quad (22)$$

where $T_{f,n}$ represents the air temperature on the n side of the battery, $T_{f,in}$ represents the air inlet temperature of the orifice plate, $T_{f,out}$ represents the air outlet temperature of the orifice plate, and $T_{c,i}$ represents the temperature of the i th cell. The transient heat transfer equation on the air side of the equivalent pipe [22] is as follows:

$$m_{air} c_{air} \frac{dT_{f,n}}{d\tau} = h_n A (\Delta T_{i-1,n} + \Delta T_{i,n}) \quad (23)$$

3.3. Battery equilibrium model

Due to the different battery capacity, internal resistance and self-discharge conditions, the battery is inconsistent. Overcharging and overdischarging may occur in the battery pack during charging and discharging. This will seriously damage the power battery life and increase the cost of battery replacement. Equilibrium management is usually used to control the voltage or quantity of the battery, to eliminate the inconsistencies in the battery pack as much as possible. In this

study, in order to better simulate the battery pack charging process on the impact of cell, dissipative passive equalization was adopted to control the voltage, and the relevant circuit schematic is shown in Fig. 7. The battery pack comprises 12 single cells in series to provide a high output voltage. Each single cell has an equalizer switch [42] driven by a PWM signal and an equalizer control unit with a corresponding equalizer resistance. The maximum equalizer ability of a single equalizer control unit is as follows:

$$i_{er,max} = \frac{U_i}{R} \quad (24)$$

where $i_{er,max}$ is the maximum equalizing current of a single equalizing unit, U_i is the voltage loaded on the i th battery, and R is the equalizing resistance. In the equalization circuit, the voltage of the lowest cell is taken as the equalization reference voltage:

$$U_R = \min_{0 \leq i \leq 12} (U_i) \quad (25)$$

The duty ratio of PWM signal control is as follows:

$$S_i = \begin{cases} 1 & U_i > U_R + U_T \\ \frac{U_i - U_R}{U_T} & U_R < U_i < U_R + U_T \\ 0 & U_i < U_R \end{cases} \quad (26)$$

where U_T is the threshold of equalization control. The charging current of the i th single battery in the battery pack can be expressed as

$$I_{ec,i} = \begin{cases} I_i(t) - \frac{U_i}{R} & U_i > U_R + U_T \\ I_i(t) - \frac{(U_i - U_R)U_i}{U_T R} & U_R < U_i < U_R + U_T \\ I_i(t) & U_i < U_R \end{cases} \quad (27)$$

To study the influence of different initial capacity and initial SOC on the equilibrium effect of battery pack. The cell capacity was randomly selected, ranging from 3.25 to 3.35 Ah. The initial SOC of the cells was

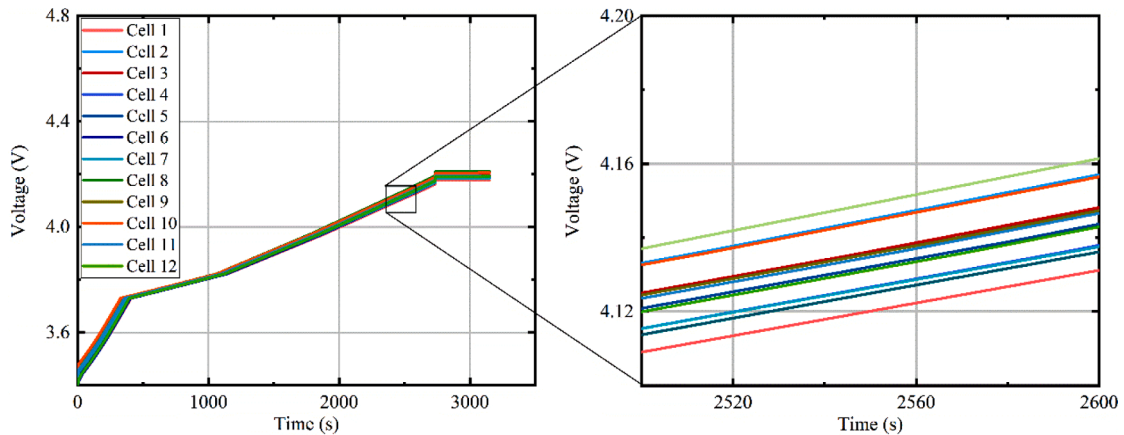


Fig. 8. Cell charging voltages without equalization circuit.

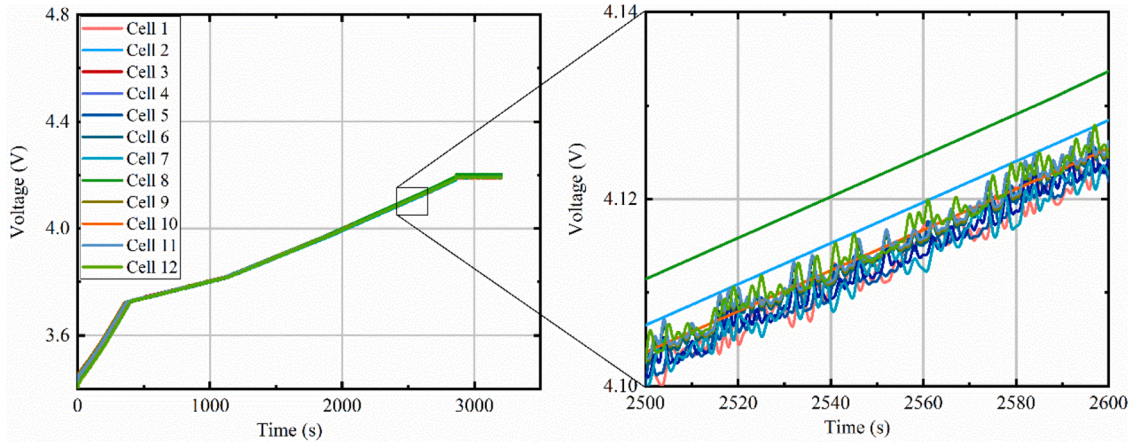


Fig. 9. Cell charging voltages with equalization circuit.

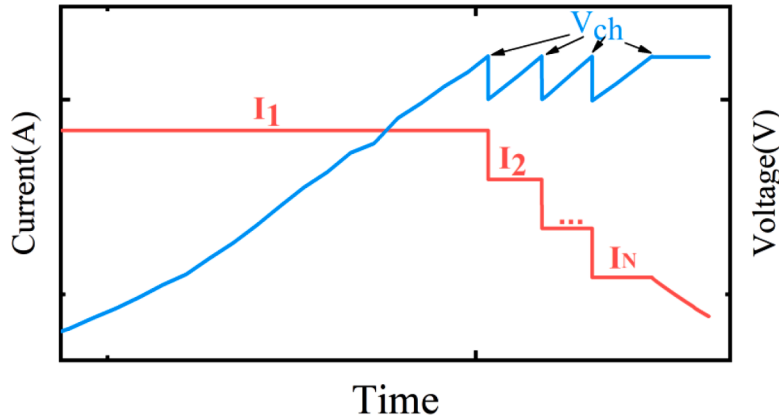


Fig. 10. Schematic of the multistage constant-current charging strategy.

randomly set from 9% to 11%. Table 3 shows the detailed capacity and SOC distribution. The average initial volume of the cells is 0.33 Ah. Taking 1C CCCV as an example, the charging voltage waveform of each single battery in the battery pack is shown in Figs. 8 and 9, in the absence of equalizing control and even control, respectively.

4. Control strategy for battery pack charging system

4.1. Adaptive MCCC charging strategy

In this study, the multi-phase constant-current constant-voltage (MCCC) charging strategy was optimized, as shown in Fig. 10. The charging profile can be divided into several CC phases and a CV phase. It is proposed [43] that multi-phase constant current is followed by a short-time constant-voltage mode to reduce the current, thereby not causing serious decomposition of electrolyte and extending the cycle life of the battery. The switching condition of CC phase is when the terminal voltage of the battery reaches the preset cut-off voltage V_{ch} . It is worth noting that the value of CC phase decreases step by step, and the gradual reduction mechanism makes it easier for the charging strategy to balance the chemical reaction and inhibit the temperature rise of the battery [44]. The CV phase is reached when the SOC exceeds 90% and the terminal voltage of the battery reaches the preset cut-off voltage V_{ch} . The different current combinations will produce different charging curves. If the battery is charged with a small current, it will take longer time to reach the cut-off voltage due to the small current, leading to fewer orders of the CC segment or even only one CC phase. On the contrary, if multi-phase constant-current charging is performed with a

large current, the battery can reach the cut-off voltage quickly, and an MCCC charging strategy involving multiple CC phases can be obtained. The CC phase value considerably influences the charging strategy. This strategy is called the adaptive-MCCC (A-MCCC) charging strategy; the CC order of this strategy is determined by the CC phase size.

4.2. Optimization objectives and constraints

Charging time and battery life are important factors affecting battery charging. Rapid charging leads to higher temperature rise, leading to reduced battery life, which is a pair of contradictory goals. Another key factor affecting battery charging is energy loss, higher energy loss will lead to lower charging efficiency. Therefore, battery charging time, battery life, and energy loss should be considered in the objective function of optimizing charging process. The complete optimization

Table 4

Physical limits and initial parameters of the battery.

State parameter	Values	Units
I	[0, 1C]	A
U_i	[2.5, 4.2]	V
T_c	[10, 45]	°C
SOH	[10, 95]	%
t_f	[0, 10]	hour
T_f	27	°C
R	40	Ω
P_{fan}	1.19	W
V_{fan}	6	m/s

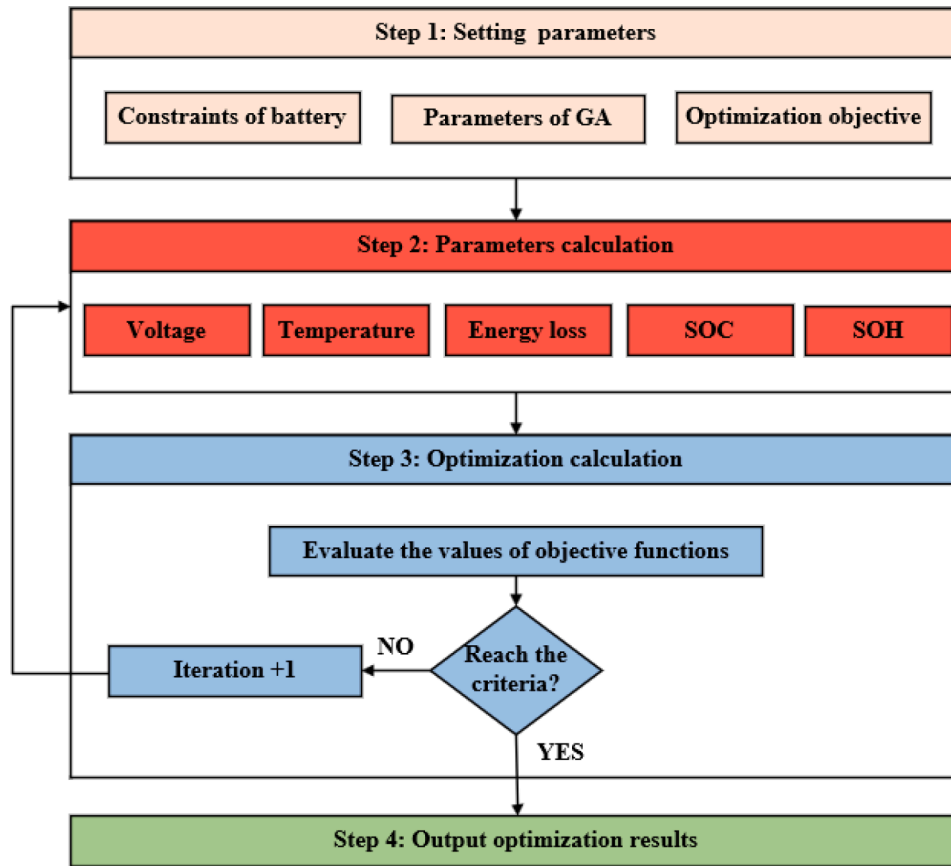


Fig. 11. Flowchart of the optimization of the model implementation.

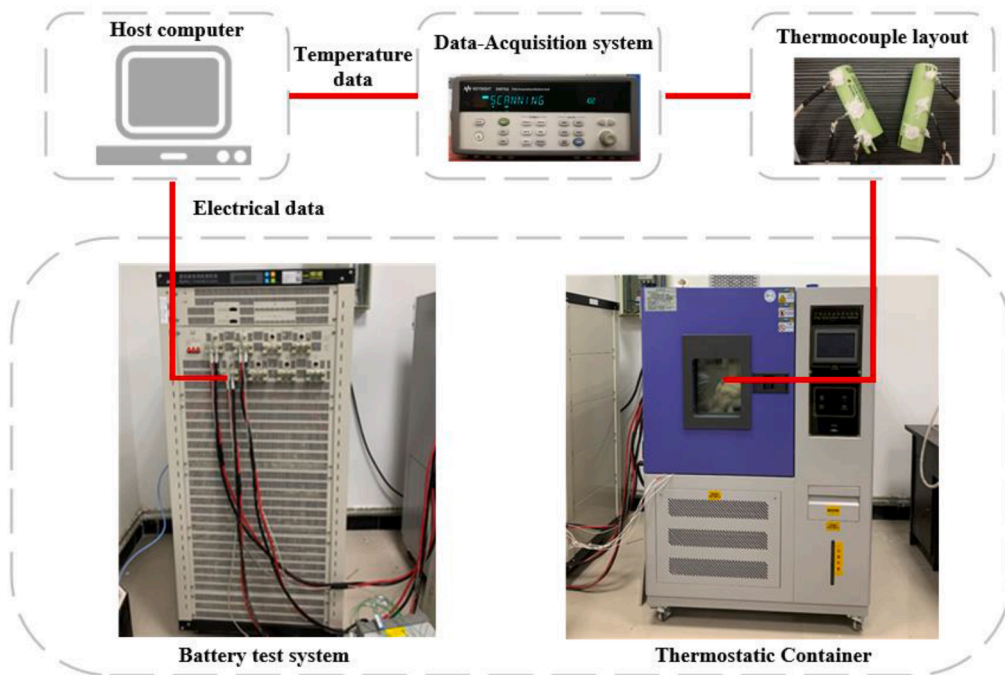


Fig. 12. Hardware setup for the battery experiment.

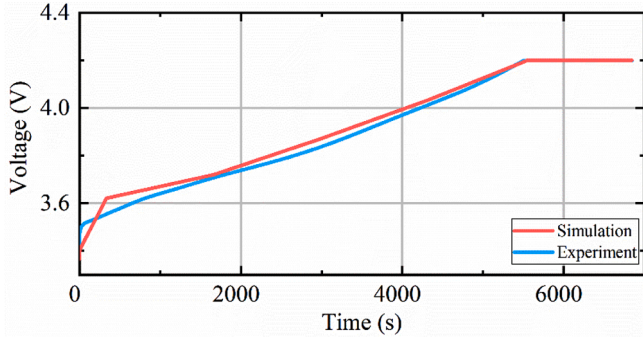


Fig. 13. Comparison of voltage response between the simulation and the experiment.

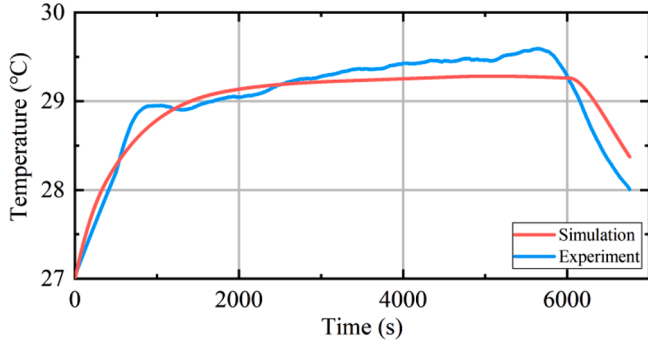


Fig. 14. Comparison of temperature between the simulation and the experiment.

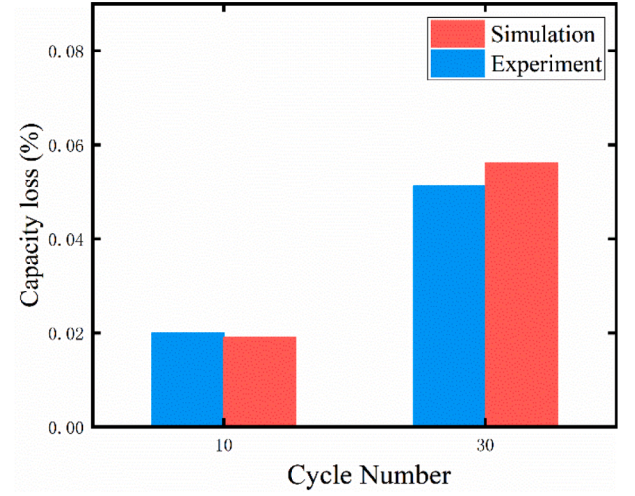


Fig. 15. Comparison of capacity loss between the simulation and the experiment.

objective function can be expressed as

$$\min J = \alpha \frac{t_f - t_0}{t_{\max} - t_0} + \beta \frac{SOH_{\text{pack}-t_f} - SOH_{\text{pack}-0}}{SOH_{\text{pack}-\max} - SOH_{\text{pack}-0}} + \gamma \frac{E_{t_f} - E_0}{E_{\max} - E_0} \quad (28)$$

$$\alpha + \beta + \gamma = 1 \quad (29)$$

The total energy loss includes energy loss when charging the battery $E_{i,c}$, energy loss due to equilibrium management $E_{i,b}$, and energy loss due to the fan E_{fan} :

$$E_{t_f} = \sum_{0 \leq i \leq 12} E_{i,c} + \sum_{0 \leq i \leq 12} E_{i,b} + E_{fan} \quad (30)$$

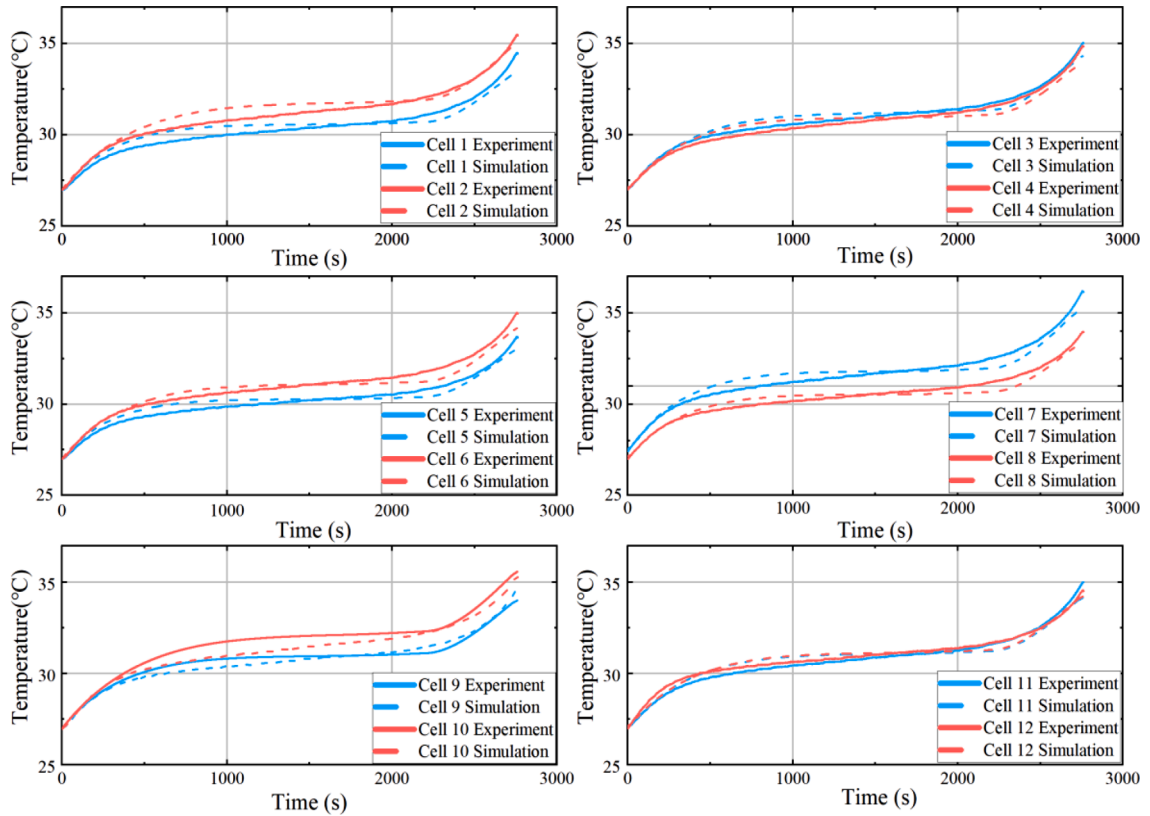


Fig. 16. Comparison of cell's temperature in battery pack between simulation and experiment.

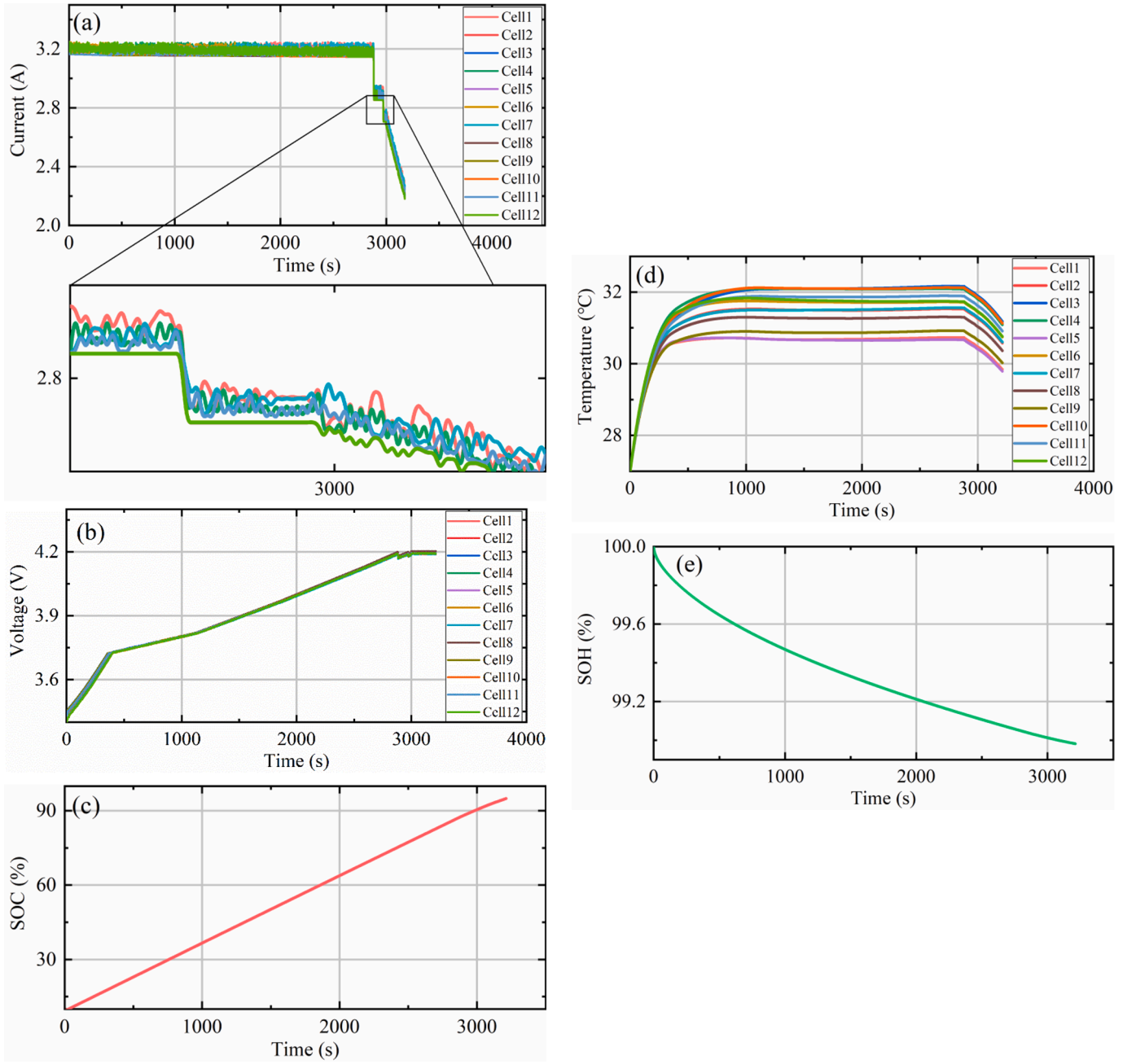


Fig. 17. Optimization results for minimum-time charging strategy: (a) current, (b) terminal voltage, (c) SOC (d) temperatures, and (e) SOH.

$$E_{i,c} = U_i I_{ec,i} t_f \quad (31)$$

$$\begin{cases} E_{i,b} = \left(\frac{U_i}{R}\right)^2 R t_f & U_i > U_R + U_T \\ E_{i,b} = \left(\frac{(U_i - U_R)U_i}{U_T R}\right)^2 R t_f & U_R < U_i < U_R + U_T \\ E_{i,b} = 0 & U_i < U_R \end{cases} \quad (32)$$

$$E_{fan} = P_{fan} t_f \quad (33)$$

To ensure the security and stability of the battery pack system, the relevant state parameters of the charging process are initially defined and constrained. Table 4 shows the specific physical boundaries and initial parameter specifications.

4.3. Optimization procedure

In this study, the value of the CC phase is taken as the optimization variable, and the charging time, battery life, and energy loss are taken as the optimization objectives. If each CC phase has N optional current values, there will be N^5 combinations throughout the charging stage (CC phase is at most 5). Clearly, testing every combination to filter the best for such a large number of combinations is not possible. A genetic algorithm is an effective method to solve constrained optimization problems. Its principle is based on the theory of biological evolution in nature. Under the rules of nature's survival of the fittest, the population evolves in a certain direction and passes on the appropriate genes to the next generation. After multiple generations of gene crossing and gene variation, the individuals that adapt to nature are finally evolved. The flowchart of charging optimization using the genetic algorithm is shown in Fig. 11. The specific steps are as follows:

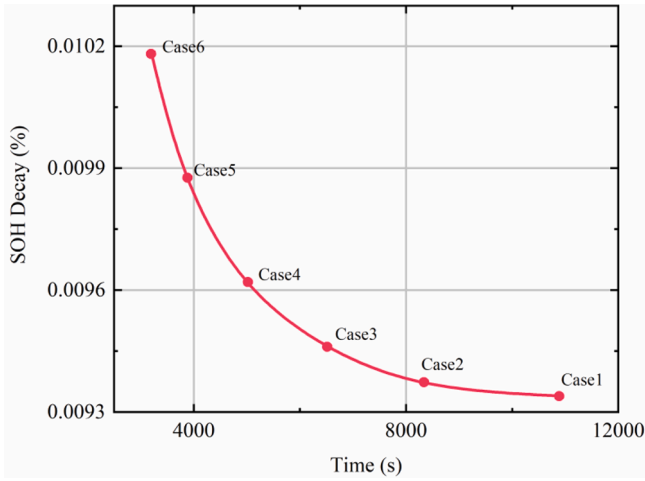


Fig. 18. Pareto frontier: charge time versus SOH decay.

Step 1: Set the battery constraints and initial parameters. Set the weight coefficient of the optimization goal. Set the parameters of the optimization algorithm—an appropriate generation number and population size—and initialize the population.

Step 2: Substitute the initial current into the model for calculating and obtaining parameters such as voltage, temperature, SOC, SOH, and energy loss.

Step 3: Calculate the value of objective function $\min J$ according to the setting of weight coefficient. If the population is selected, crossed, recombined, and mutated to meet the requirements to form a new population, return to step 2.

Step 4: Meet the termination conditions and output the optimization results.

5. Results and discussion

This section presents the results of multi-objective optimization charging. First, the minimum-time charging strategy is optimized, and the optimized charging strategy is analyzed and explained. Second, without considering the energy loss, the contradiction between charging time and battery life is studied, and the corresponding Pareto boundary diagram is analyzed in detail. As battery aging and energy loss are not strongly correlated targets, the interaction between charging time and energy loss is investigated. Finally, a balanced charging strategy considering charging time, battery life, and energy loss is proposed.

5.1. Battery model experiment verification

To verify the accuracy of the battery model used in this study, related experiments were performed on the NCR18650B battery. The experiment setup comprises (1) a battery test system (Neware BTS4000), (2) a data collection system (Keysight 34970A) with a K-type thermocouple, (3) an incubator (CK-150G), and (4) a cylindrical battery (NCR18650B). The experimental equipment is shown in Fig. 12. To verify the accuracy of the coupling model, a 0.5C CCCV method was used for the charging experiment, and the analog voltage, surface temperature, and capacity loss were recorded. Comparisons between the simulation model and experimental results are shown in Figs. 13–15. The root-mean-square error of voltage is 70 mV, and the root-mean-square error of temperature is 0.3 °C. The relative error of capacity loss is 8.6%. To sum it up, the relevant model of the battery herein can better reflect the actual physical process of the battery. But the capacity loss is more accurate in the first 10 cycles. As the number of cycles increases, the model error increases because the electrical parameters of the battery had a longer timescale than the SOC and temperature. The electrical parameters of the battery change with the increase of the cycle, especially the increase of the

internal resistance of the battery, which causes the battery to reach the full state early, which makes the amount of charge less.

The battery pack thermal model was validated, and the fan was used as the external power to improve the cooling air during the test. The experiment was performed using a 1C constant discharge, and the simulation results of the temperature were compared with the experimental results, as shown in Fig. 16. The proposed simulation model can well simulate the cooling behavior of the battery pack.

5.2. Minimum-time charging strategy

The shortest charging time is sought. The objective function contains only the charging time ($\alpha=1$, $\beta=\gamma=0$):

$$\min J = \alpha \frac{t_f - t_0}{t_{\max} - t_0} \quad (34)$$

The genetic algorithm optimization results are shown in Fig. 17. To ensure that the charge can be found within a shorter period filling the more power, the entire process is divided into three different stages. The first stage is the maximum constraint for charging current, the charging current in load voltage increase in the ohmic resistance and the polarization resistance makes the battery voltage rising fast. When the cut-off voltage is reached, the second stage begins, and the charging current is reduced. The third stage continues reducing the charging current when the voltage reaches the cut-off voltage again. The purpose is to reduce the battery temperature and reduce the loss of battery life while ensuring rapid charging. At the SOC of 90.34%, it enters the fourth stage, charging in the constant-voltage mode, thereby ensuring full power. The entire charge took 3212 s. The highest battery temperature rises to 32.2 °C. The charging energy loss is 134.7611 Wh. Owing to the effect of equilibrium management, the current of each battery in the battery pack fluctuates all the time, and the fluctuation range is different. Compared with the charging strategy for a single battery, the battery pack charging strategy places greater importance on the parameters of the status of the battery pack, such as SOC and SOH; moreover the requirements for a single battery are limited to ensure that the safe range is not exceeded. As a result, the optimization of battery pack charging strategy becomes more complicated. All the charging optimization takes battery pack parameters as the priority to ensure the safety of single battery through equilibrium management. In addition, owing to composition of the battery pack, the temperature rise of single batteries is different, although the trend is consistent, as shown in Fig. 17 (d); this also increases the difficulty of charging optimization. In the optimization of the shortest charging time, the objective function does not include aging and energy loss. Therefore, the minimum-time charging strategy resulted in large energy loss and serious aging.

5.3. Contradiction between charging time and battery life

Battery life is another important factor for evaluating the charging process as it directly affects the performance of the battery. We include the battery life in the target function. In addition, charging time and life are two contradictory parameters [45]. The Pareto relation between charging time and lifetime was obtained via genetic algorithm optimization. The objective function is as follows:

$$\min J = \alpha \frac{t_f - t_0}{t_{\max} - t_0} + \beta \frac{SOH_{t_f} - SOH_0}{SOH_{\max} - SOH_0} \quad (35)$$

The Pareto boundaries of charging time and life are shown in Fig. 18. The left area of the figure shows that the battery charging time decreases rapidly and the aging increases less. Faster charging can be achieved at the expense of slightly accelerating the battery aging. Fig. 19(a) shows the optimization of charging current with different weight coefficients. Under different weights, the current value of CC phase and the number of CC phase are different, reflecting the characteristics of self-adaptive MCCC charging strategy. With the increase in β , the optimization

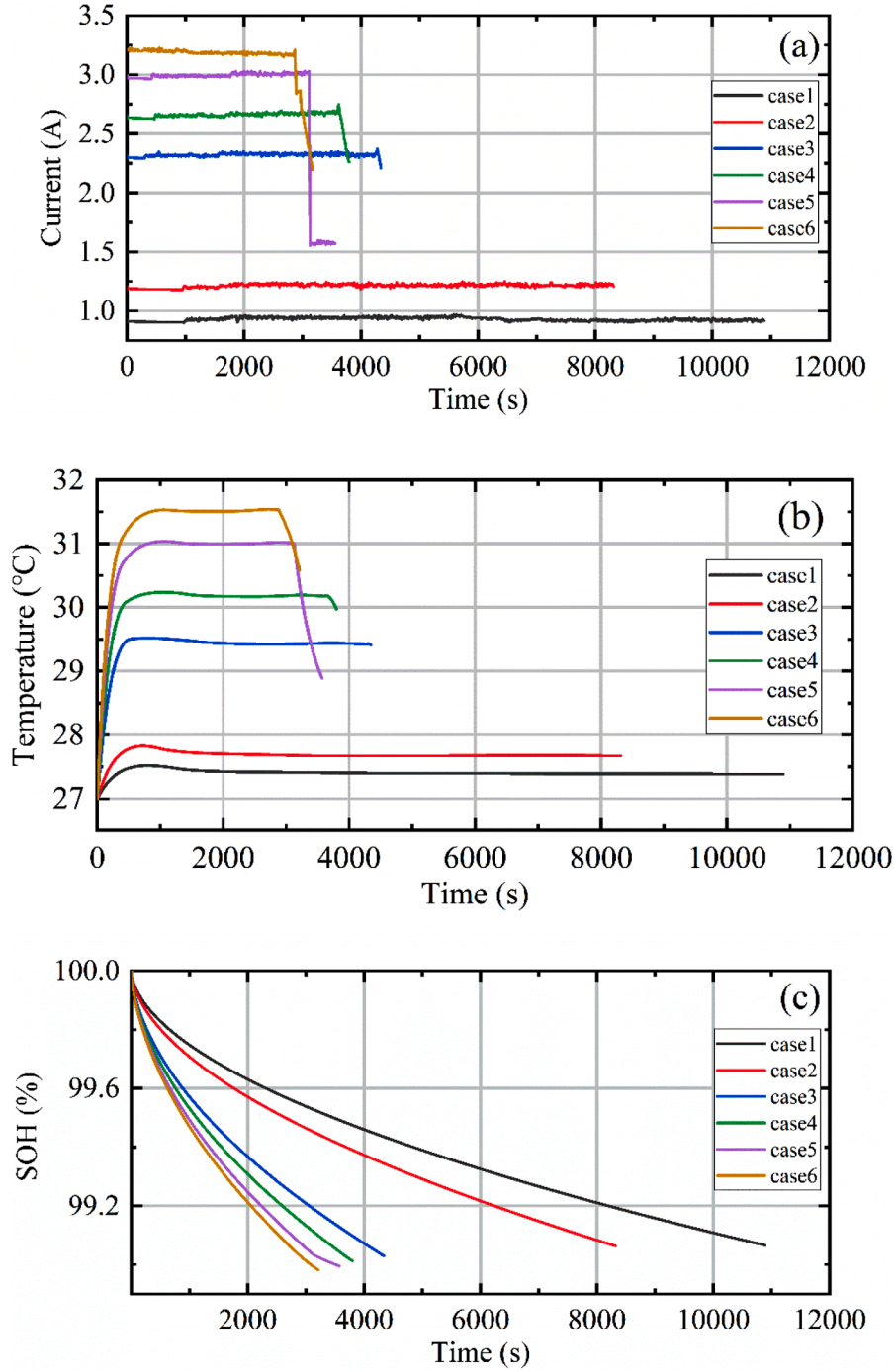


Fig. 19. Charging strategy considering charging time and SOH. (a) Optimized charging current profiles at different weighting factors. (b) Temperature rise at different weighting factors. (c) SOH at different weighting factors.

focuses more on the battery life, and the battery temperature has a significant influence on the battery life. Thus, avoiding the large current charging in the initial stage is necessary. This is because the battery in low SOC has a large internal resistance, and the loading of a large current charging leads to an increase in battery heat generation, which in turn leads to the rise in battery temperature.

5.4. Interplay between charging time and charging loss

Reducing charge loss is crucial to improve charging efficiency. In the process of single-battery charging, charging time and charging energy consumption conflict with each other [11]. However, in this paper, the

energy loss includes not only the energy loss of charging but also the energy loss of equilibrium management and the power loss of external fans. Therefore, studying the correlation between battery pack charging time and energy loss is important. Here, we extend the cost function of the shortest charging time in Section 5.2 to include the total energy loss. The objective function with the weight factor is as follows:

$$\min J = \alpha \frac{t_f - t_0}{t_{\max} - t_0} + \gamma \frac{E_{t_f} - E_0}{E_{\max} - E_0} \quad (36)$$

The Pareto boundary of the charging time and energy loss is shown in Fig. 20. The γ values of cases 1–6 are 1;0.99;0.95;0.9;0.85;0. As shown in Fig. 21, the charging energy accounts for a large part of the energy loss

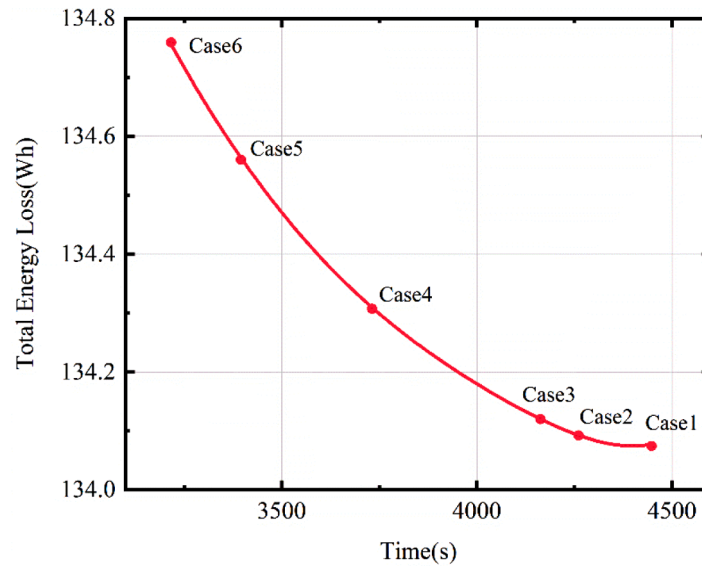


Fig. 20. Pareto frontier: charge time versus charging loss.

of charging. As an additional loss in charge management, energy loss by equilibrium management and energy loss by the fan is small. With the increase in γ , the optimization focuses more on the total energy loss; to achieve this goal, the battery current is limited. This is because an excessive charging current leads to a large energy loss during charging, and the energy loss during charging accounts for a large proportion, thus considerably affecting the total energy loss. With the decrease in charging current, the proportion of $E_{i,b}$ and E_{fan} increases, but the total energy loss decreases. However, when only total energy loss is considered ($\gamma=1$), the current is 2.33 A, and it does not continue being downregulated. When the current was less than 2.33 A, the energy consumed increased instead. This is because the equilibrium management and fan are always working, and the longer the charging time, the higher is the energy consumed. In addition, as shown in Fig. 20, after Case 3, the energy loss can be substantially reduced by reducing the charging time; in contrast, the impact before Case 3 is relatively small. To minimize the charging time while minimizing the energy loss, the charging strategy for the left area of Fig. 20 should be considered as much as possible.

5.5. Balanced charging strategy with three objectives

Battery aging and energy loss are important factors that affect charging. These two factors are added to the objective function to optimize the charging strategy and achieve a balanced charging strategy that considers charging time, aging, and charging energy loss. Fig. 22 shows the optimization results of three-target charging in the form of Pareto boundary. As there is not much relation between battery aging and energy loss, the Pareto curve of three-objective optimization presents the curve form in three-dimensional coordinates. Sections 5.3 and 5.4 respectively provide the optimal areas for charging time and aging, charging time, and energy loss. The optimal charging current can also be obtained according to this law in the Pareto diagram of three-objective charging.

The weight coefficients of the balanced charging strategy are $\alpha = 0.2$, $\beta = 0.03$, and $\gamma = 0.77$. The optimized charging current trajectory is shown in Fig. 23. Balanced charging strategy has lower energy loss and aging than fast charging strategy. However, it takes an extra 440 s to reach its charging target. Although the average initial CC phase current reached 2.82 A, there was only one CC phase. This was because the voltage of the battery was constantly fluctuating due to the effect of equilibrium management; thus, the battery voltage reached the cut-off voltage in advance, and the number of CC phases was reduced. As

shown in Fig. 24, the monomer battery recharged with the same average initial CC phase, and a second CC phase appears. In addition calculation formula according to the SOC of battery pack can be found that the battery pack SOC is the ratio of the minimum available capacity of a monomer battery to the minimum capacity of a monomer battery in the battery pack., which makes SOC of battery pack is greater than or equal to the largest single SOC of the battery.

6. Conclusion

In this paper, we first established a coupled single-battery electric heating aging model and then added the battery pack cooling model, and the battery pack balanced management model formed a complete battery pack model. Our proposed model can fully describe the state parameters of the battery pack and the interaction of the single battery in the battery pack in terms of electrothermal aging. The cooling model of the battery pack can well describe the temperature inconsistency caused by the difference of heat dissipation caused by the different positions of the battery pack. The equilibrium model can well describe the electrical inconsistencies caused by the manufacturing process and material of the cell in the battery pack. The complete battery pack model is helpful to study the influence of battery pack charging process on single battery. A multi-objective optimization framework was developed to realize an optimal charging strategy. Under this framework, battery equalization, temperature and battery parameters are limited, and the optimization targets are charging time, aging, and charging energy loss. Then, a GA method is used to optimize the A-MCCCV charging strategy. According to the studies, the three important conclusions are achieved:

1. The rapid charging strategy with charging time as the only target has faster charging speed but greater aging and energy loss. The contradictory relation between charging time and aging is revealed, and it is suggested that the balanced charging strategy should be carried out near the left side of the Pareto curve.
2. Our study also observed a contradictory relation between charging time and energy loss; however, it is not that the longer the charging time, the lower the energy loss is. Because the balance management and the fan are always working, the longer the charging time, the higher is the energy consumed.
3. A balanced charging strategy that considers the three targets is obtained. Compared with the single battery charging with the same average current, the number of CC phase decreases, which is owing to the influence of balanced management and battery pack SOC.

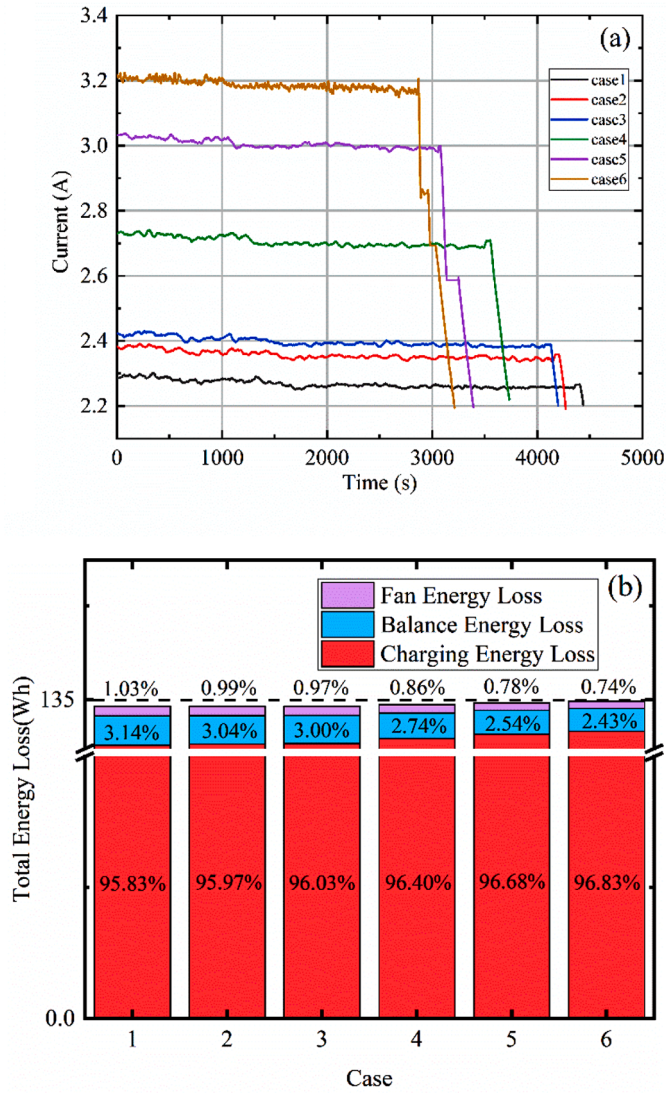


Fig. 21. Charging strategy considering charging time and charging loss. (a) Optimized charging current profiles at different weighting factors. (b) Charging loss at different weighting factors.

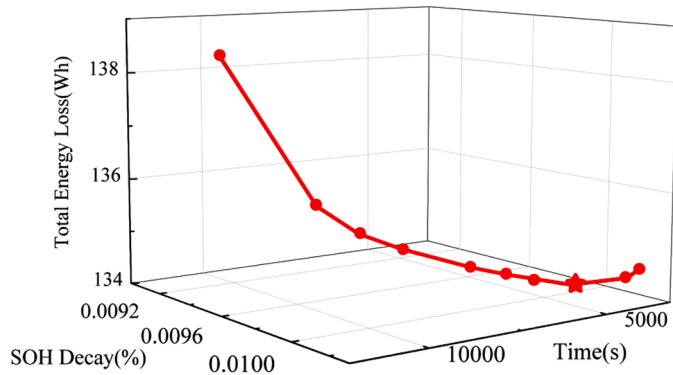


Fig. 22. Pareto frontier, charge time; SOH decay versus charging loss.

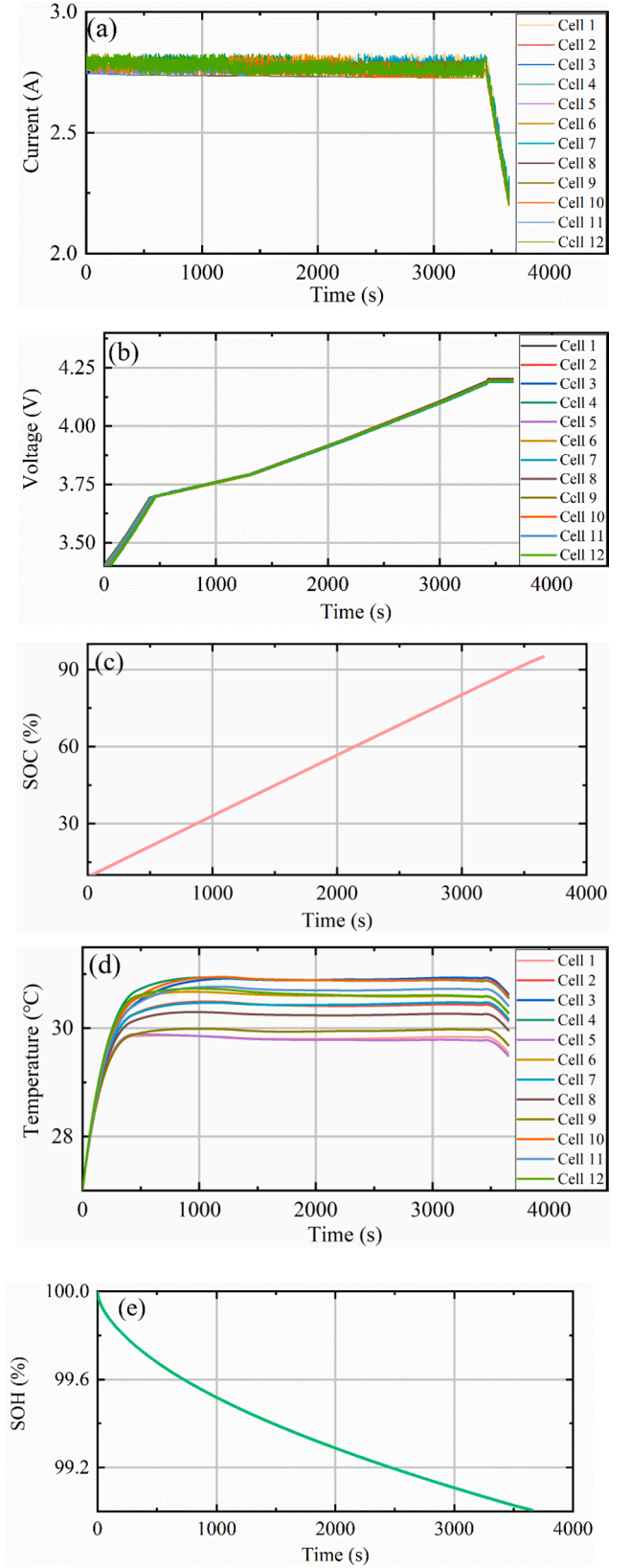


Fig. 23. Optimization results for balance charging strategy: (a) current, (b) terminal voltage, (c) SOC, (d) temperatures, and (e) SOH.

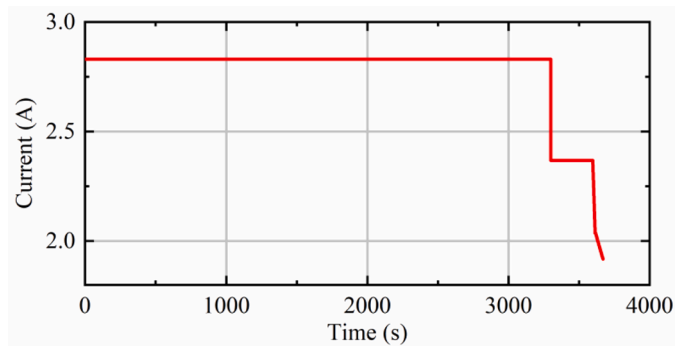


Fig. 24. Current of cell by balanced charging strategy.

CRedit authorship contribution statement

Yunjian Li: Methodology, Writing – original draft. **Kuining Li:** Conceptualization, Funding acquisition. **Yi Xie:** Conceptualization, Writing – review & editing. **Bin Liu:** Resources, Project administration. **Jiangyan Liu:** Writing – review & editing. **Jintao Zheng:** Validation, Data curation. **Wei Li:** Validation.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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